Mapping Geo-Spatial Data Transcript

Now, we get into a very interesting category of data mapping geospatial data, it is my personal favourite where because we live in a real time space and experience the spatial phenomenon. However, plotting spatial data requires special tools and special understanding of the data structure. We are not getting into the technicalities of mapping geospatial data, we are using the spatial algorithms or the spatial visualisations to present the quantitative or qualitative data on a spatial map and that can better convince the audience because they are living in a space which is related to the map.

For example, whenever we see a map of a nation and a phenomenon that is plotted over it, it is easy for us to correlate than seeing it as a bar graph or as a scatter plot. The very fact it is presented on a map gives a closure and relation to the audience naturally grabbing their attention and providing another layer of comfort and acceptance. First of all, the spatial data is plotted on chloropleth map.

So, we will now look at mapping spatial data. Spatial data are very interesting in their own right, they relate to this living space in which we inhabit and they are easy to relate for the audience. Whenever you present a phenomenon on a spatial map, it is easy to get attached and interpret a meaning to it than it is presented in case of a bar or a scatter plot.

The relation that a geographical space brings is immense and can create connect with the audience when that is why the increasing there is an increased use of spatial representation of data using map coordinates and representing them as spatial visualisations. The basic spatial visualisation is a chloropleth map. Here it uses colour saturation to distinguish between the quantitative or qualitative difference in the variable that we want to present over the reading.

The example that we are showing here is the annual unemployment rate of the US. In 2004 it is around 5.5 percent, in September 2009 around the global financial crisis period it has reached 9.8 percent, almost doubling close to doubling and you can see the regions where the effect is more ah severe. However, one of the problems with using chloropleth map is it may not be ah proportionately representative of the population data.

We will now look at mapping spatial data using chloropleth maps. Chloropleth maps use colour saturation to present the qualitative data or quantitative data over a geographic region. This can be used to distinguish between the levels of changes that are occurring between two geographical regions in a given time period and the intensity of the colour represents the intensity of the variable that we are looking at.

For example, in the graphic that you are seeing on your slide the unemployment rate of US in 2004 versus 2009 is presented. The regions which are darker in colour on the 2009 map indicate the places where the unemployment is higher. However, there is a small catch or a problem with the chloropleth maps.

The intensity might not be true representation of the population data. The example that you are seeing showcases the unemployment of US between 2004 and 2009. So, care has to be taken to understand the nature of data in relation to the population size and the geographical area and if needed some normalisation exercise has to be done to ensure the colour saturation truly represents the phenomenon as per the intensity.

This is something we have to take care while using the spatial data and presenting them using spatial visualisations like chloropleth map. To address this we have isothermic map which takes care of the problem which is associated with the chloropleth map and this algorithm helps to correct for the underlying geographical area and it is a population and it tries to address the colour saturation and present the visual in a more coherent fashion representing the actual phenomenon. So, these options are available in advanced tools and has to be chosen carefully for presenting the data.

And we have network connection map which is also another form of geospatial representation which tries to represent the connections between 2 or more geographical areas and forms a closure. Again following just like principles even though the maps are not drawn because the connections form a closed figure or provide closure it is easy to see they are forming a kind of a map or in this case presenting the world map. Likewise whenever you do a network map connection it might be the case that the networks can provide the closure of the underlying maps and directly provide the map or you can use a map on which the networks can be plotted.

I am again emphasising the fact the spatial visuals that we are looking at this lesson can be used or intended to be used for custom visualisations and they have to be procured or processed in special softwares which we will be looking in our experiential learning sessions.

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